

SAGNIK CHATTERJEE

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EDUCATION

University of Massachusetts Amherst

MS, Computer Science

Amherst, MA

2025 – 2027

- Coursework: Advanced NLP, Software Engineering, AI Alignment, Trustworthy AI

Manipal Institute of Technology

BE, Computer Science and Engineering

Manipal, India

2018 – 2022

RESEARCH EXPERIENCE

SAGE Lab, University of Massachusetts Amherst

Graduate Student Researcher, Advisors: Saishradha Mohanty, Anna Green

Amherst, MA

Jun 2026 – Aug 2026

- Engineered **cross-layer transcoder (CLT)** training and evaluation pipelines for ESM2-650M, integrating activation extraction, sparse feature learning, and ProteinGym mutation-effect evaluation.
- Profiled and optimized GPU workloads on the UMass **Unity cluster** via distributed execution, memory-efficient processing, and inference-pipeline improvements.

BioNLP Lab, University of Massachusetts Amherst

Research Assistant, Advisors: Sunjae Kwon, Hong Yu

Amherst, MA

Sept 2025 – Dec 2025

- Fine-tuned **LLaMA-3.2-3B-Instruct** with LoRA on the SNOMED CT and MeSH ontologies using the **OntoTune** self-training framework, gaining **+6.25 pp** accuracy over the zero-shot baseline on MedMCQA and USMLE.
- Designed an ontology-driven **dual-agent self-play framework** where a MeSH/PubMed-grounded question agent and a text-only answer agent co-evolve via **DPO** with adaptive difficulty scheduling, targeting the model's biomedical knowledge gaps without external knowledge access at inference time.

PUBLICATIONS

Reddy, V.V., **Chatterjee, S.**, Bhatta, S. *Selecting Worker-Agent Training Context Under Continual Tool-Use Shift Managing Agents that Manage Agents Workshop, NeurIPS'2026. **Under Review.***

Maniyar, S., Mishra, A., & **Chatterjee, S.** *Is the Refusal Direction Mostly Harmfulness? A Multi-Model, Multi-Scale Reproduction of "LLMs Encode Harmfulness and Refusal Separately"*. Transactions on Machine Learning Research (TMLR). **Under Review.**

Chatterjee, S. *Refusal Steering using SAEs* New England Mechanistic Interpretability (NEMI) Workshop 2026. **Accepted.**

PROFESSIONAL EXPERIENCE

IBM India Pvt. Ltd.

Software Engineer

Gurgaon, India

Sept 2022 – Jul 2025

- Tasked with reducing manual triage on an Indian Oil Corporation (IOCL) support queue of $\sim 1,200$ tickets/month; designed a **Python-based NLP automation system** pairing TF-IDF and logistic-regression classification with rule-based resolution, **cutting average classification time from 28–30 to 6–8 minutes** across a 350-ticket pilot at $\sim 78\%$ accuracy.
- With triage logic spread across disconnected enterprise tools, developed modular backend services and **REST APIs** for ticket ingestion, ML inference, knowledge-base retrieval, and root-cause analysis, containerized with Docker, **raising SLA compliance from 78% to 92%** through API and observability improvements.
- Because incorrect auto-resolution was unacceptable in a compliance-sensitive production environment, implemented confidence-based routing, human-review fallback, and auditable decision logging, making every automated decision traceable and clearing the system for production deployment.

PROJECTS

anchora | Durable Workflow Engine for LLM Agents | GitHub

Go, PostgreSQL, Redis, chi, Eino, Docker

- Built a **DAG workflow engine** in Go that validates agent step graphs, rejecting cycles and duplicate, self, or unknown dependencies, then runs each ready wave concurrently, templating upstream outputs into downstream prompts.
- Designed a durable **at-least-once job queue** on **PostgreSQL and Redis** where workers claim jobs into a deadline-scored lease set and renew by heartbeat, with paired Redis-lease and PostgreSQL-orphan reapers requeueing work stranded by dead nodes.
- Fenced worker writes on lease ownership** so a partitioned worker cannot overwrite its replacement; redelivered jobs resume by replaying succeeded steps from the store rather than re-invoking agents, and jobs past a delivery cap are dead-lettered.
- Exposed a **chi HTTP API** with synchronous and job-based execution, resumable **Server-Sent Events**, and a cluster endpoint for queue depth and worker roster.

TECHNICAL SKILLS

Languages: Python, Go, C, Javascript, SQL

Agents, Post Training and Interpretability: SmolAgents, PyTorch, TransformerLens, LoRA/QLoRA, SFT, DPO

Backend & Distributed Systems: FastAPI, PostgreSQL, Redis, Kafka, REST APIs, Docker, Kubernetes

Cloud & Developer Tools: AWS, Git, PyTest, Claude Code

CERTIFICATIONS

Technical AI Safety Project (BlueDot, 2026) | **Technical AI Safety** (BlueDot, 2026) | **AWS Cloud Practitioner** (AWS)